

Information loss of fixed linear health indicators as a control problem: simplex geometry and reachability limits

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Abstract

Fixed-weight health indicators discard information: the channel combination most informative about remaining life rotates as degradation mechanisms progress at different rates. We show the operating point controls how much is lost. For a linear indicator, positive degradation rates and Gaussian remaining life over a time-shift population, the loss depends on the indicator only through one misalignment angle, a distance on a statistical simplex. Zero loss is attainable exactly when the informative directions reachable at each instant share a common point, which generically needs one fewer independent operating variable than there are mechanisms. Invariant to common channel scaling, the loss is invisible to every criterion built from the Fisher information about remaining life. On a representative synthetic plant, steering lowers the information-loss objective 68-fold against the best constant operating point. Across 299 random plants of the separable model, the reachable-set width, known before design, provides a moderate screening signal for when steering pays (area under the receiver-operating curve 0.73). In six public fleets, measured battery cells show the rotation and simulated turbofan records, checked against a negative control, support its dependence on the operating point; the highest-fidelity turbofan data reject that separable form.

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1. Introduction

Condition monitoring rarely feeds raw measurements to a life-prediction algorithm. A vibration channel, a temperature, an impedance and a capacity reading first go into a single scalar, the health indicator, and the life-prediction algorithm sees only that scalar [1, 2]. The compression is almost always fixed: the designer picks one set of weights at design time, and every unit carries those weights for its whole life, because firmware embeds them, certification freezes them, or re-tuning them in the field is impractical.

Fixing the compression has a cost. Real components degrade through several mechanisms at once, and those mechanisms do not proceed in step: an early-life process may dominate for the first third of the record and then saturate, while a late-life process accelerates. The combination of channels that carries the most information about remaining life therefore rotates as the component ages. A fixed indicator is aligned with that combination at best at one moment, and is misaligned for the rest of the record; the information it discards is not recoverable downstream, however good the prediction algorithm that follows [3].

The established response is to make the map better: choose a different criterion for it, let it switch between segments of life, or learn it from more run-to-failure data [4, 2]. All of these take the degradation as given and work on the compression. This paper acts on the degradation itself, through the operating point.

The rotation is not a property of the instrumentation. The degradation physics produces it, and that physics responds to stress: temperature, load and duty cycle change the relative rates of the competing mechanisms, not only the overall speed. Running a cell hotter does not only age it faster; it also shifts the balance between the mechanisms, and therefore moves the direction in which information about remaining life arrives.

The operating point is, in most applications, something the operator chooses. So the misalignment between a fixed indicator and the informative direction is not fixed by nature. It is a function of the input, and the operator can steer it.

That reframing turns a data-analysis question into a control question, and control questions come with sharper answers than design heuristics do. One

can ask what misalignments are reachable at a given operating envelope, whether zero loss is attainable at all, how many independent inputs that would take, and what the best achievable loss is when it is not. These are the questions this paper answers.

They sit against five bodies of work. Four address parts of the problem and the fifth supplies the geometry used in Section 3; to our knowledge none of them addresses the position taken here.

The prognostics literature treats the compression as a design problem to be solved from data: searching genetically over candidate features against monotonicity and trendability criteria [2], learning it with a recurrent network [5], or surveying the resulting pipelines [1, 6]. Two limits on that programme are known, information-theoretic bounds on what any fixed compression can retain [3] and the number of run-to-failure units needed to learn one reliably [4]. The first is our own, and the boundary is worth drawing explicitly: it takes one degradation trajectory as given and asks how much information a fixed compression must lose along it. The question here is the complementary one, whether that trajectory can itself be steered so that the same fixed map loses less. Both prior limits hold the trajectory fixed; neither asks whether it can be changed.

The predictors downstream are overwhelmingly stochastic — Wiener, gamma and inverse-Gaussian degradation processes with first-passage failure [7, 8], reviewed for remaining-life estimation in [9, 10] and embedded in system-level design methodology in [11]. That literature models how damage evolves, not how the measurement should be compressed before the model is applied, which is the question upstream of it.

A separate literature does treat the operating point as a decision variable, but to extend remaining life rather than to preserve estimability [12, 13]; the two aims can conflict, and Section 6 exhibits an instance in which they disagree. Choosing inputs to make a state easier to estimate is itself classical, going back to dual control [14], and its modern forms shape a scalarisation of the Fisher information matrix [15, 16, 17]. Our Proposition 4.6 shows the quantity at issue here to be invisible to every such scalarisation.

The fifth body is information geometry. The reformulation of Section 3 rests on the Fisher–Rao geometry of the probability simplex and on the Bhattacharyya coefficient [18, 19], and the flow it produces is the replicator equation of evolutionary dynamics [20, 21]. These are standard objects; what is new here is their identification with the informative direction of a degrading component, and the fact that the fitness driving the flow is actuated.

Table 1: Where this paper sits. The columns are the axes that separate these literatures, not a ranking: each body of work answers a question the others do not ask, and the last column is a restriction this paper imposes on itself rather than an advantage over the others. “Input alters degradation” means the operating point enters the degradation dynamics; “direction” means the informative direction of Section 2 is itself an objective.

	input alters degradation	lifetime is an objective	direction is an objective	indicator fixed, scalar
Indicator design [2, 5, 1]	—	—	—	✓
Fixed-indicator limits [3, 4]	—	—	—	✓
Stochastic life models [7, 9]	—	—	—	—
Health-aware control [12, 13]	✓	✓	—	—
Observability-aware [14, 15, 17]	✓	—	partly	—
This paper	✓	secondary	✓	✓

The contributions are as follows.

1. The information a fixed indicator discards is an exact function of one misalignment angle (Lemma 2.2). Placing the channels’ shares of the Fisher information on a statistical simplex turns the degradation into a replicator flow whose rates the operating point modulates, and the optimal indicator into a barycentre of the degradation path (Section 3).
2. Zero loss is attainable exactly when the reachable direction sets share a point, which generically needs at least as many independent operating variables as mechanisms less one (Section 4). Supplying the missing input collapses the peak reachability floor from 4.52° to zero at every state, so the rule prices instrumentation rather than only bounding it. Taking logarithms makes every reachable set a translate of one fixed set, which turns the question into a containment and shows the counting rule to be sufficient but not necessary; the exceptions exist for any number of mechanisms and are too fragile to build on (Section 4.1). The count separates two objectives that are easily conflated: holding the misalignment at a constant level is a single scalar condition and generically needs one input, whereas freezing the informative direction outright is $d - 1$ conditions. That gap is the difference between useful steering and lossless steering, and an installation with one controllable variable can have the first without the second.
3. The loss is invariant to a common gain on all channels and is therefore invisible to every scalarisation of the Fisher information about remain-

- ing life, so no criterion built from that information alone can rank indicators by the loss they incur.
4. The best achievable loss is the value of an optimal control problem whose objective is a ratio, and whose terminal time is itself controlled (Section 5); we compute its value to two digits and do not certify it. The actuation authority, measured by the width of the reachable set and computable before any indicator is designed, provides a moderate screening signal for the regime in which steering can have substantial value; on 299 randomly drawn plants it ranks them as well as its ratio to the drift of the informative direction and fits them better, and the drift matters only in combination with it (Section 6.7).
 5. The premises are separately testable on public run-to-failure data, and Section 7 tests three in turn: whether the informative direction rotates, whether the operating point moves it, and whether the separable stress model survives. The first holds on measured fleets, the second so far only in simulation, and the third is rejected. That section therefore reports which hypotheses withstand contact with data rather than demonstrating that the method works.

The results do not all rest on the same assumptions, and it is worth separating them before any of them is stated. Lemma 2.2, Proposition 2.3 and Theorem 4.1 need only the observation model of Section 2: a fixed linear indicator on whitened channels, and a scalar latent. Lemma 4.3 and Proposition 4.4 need more, namely that the input enters each mechanism through a single separable factor, and Section 7 reports that the highest-fidelity turbfan data reject exactly that. The empirical readings of Section 7 need more again, since estimating one informative direction from a fleet presumes the units differ by a time shift rather than by their degradation coefficients. Fig. 6 is the map, and the reader who wants to know which branch a given result sits on should consult it rather than infer it.

Section 2 states the model. Sections 3 to 5 develop the theory, Section 6 exercises the three of its claims that are stated sharply enough to be refuted, and Section 7 reports what public fleet data can and cannot establish about the premises the theory rests on.

2. Problem formulation

This section states the degradation model, says precisely what a prognostic algorithm is being asked to estimate and where the uncertainty in that

132 quantity comes from, derives the direction along which information about
 133 it arrives, and defines the fixed indicator whose information loss the rest of
 134 the paper studies. Readers familiar with [3] will recognise the model; the
 135 derivation is given in full here because the geometry of Section 3 depends on
 136 several of its details.

137 2.1. Degradation driven by a controlled operating point

138 A component degrades through d distinct mechanisms acting at once. In
 139 a lithium-ion cell these might be loss of cyclable lithium, growth of the solid-
 140 electrolyte interphase and mechanical fatigue of the electrode; in a rolling
 141 bearing, spalling, wear and lubricant degradation. We write $x_i(t) \geq 0$ for the
 142 damage accumulated by mechanism i up to time t , and collect these into the
 143 state $x = (x_1, \dots, x_d)$.

144 One externally chosen operating point drives all d mechanisms. It collects
 145 the temperature, the mechanical or electrical load, and the duty cycle at
 146 which the operator runs the component. We write that operating point as

$$u(t) \in \mathcal{U} \subset \mathbb{R}^m, \quad (1)$$

147 with $\mathcal{U}$ compact, representing the operating envelope the application permits.
 148 Throughout this paper u is the control input: it is the decision variable,
 149 and the question we ask is what choosing it well can buy. The number of
 150 independent inputs, m , turns out to decide what is achievable, and it is
 151 usually much smaller than the number of mechanisms d .

152 The damage states evolve according to

$$\dot{x}_i = \varphi_i(u) h_i(x_i), \quad i = 1, \dots, d, \quad (2)$$

153 in which the two factors play different roles:

- 154 • $\varphi_i(u) > 0$ is the stress-acceleration factor of mechanism i : how much
 155 faster that mechanism runs at operating point u than at a reference
 156 point. It carries all the dependence on the input.
- 157 • $h_i(x_i) > 0$ is the intrinsic shape of mechanism i : how its rate depends
 158 on the damage it has already done. It carries all the dependence on
 159 the state.

160 This separation is the standard accelerated-life form and covers the stress
 161 models in routine use [22]. In the Arrhenius model the input is absolute

162 temperature and $\varphi_i(u) = A_i \exp(-E_i/(ku))$, with $A_i > 0$ a pre-exponential
 163 constant, $E_i > 0$ the activation energy of mechanism i and k Boltzmann's
 164 constant. In the inverse-power-law model the input is a mechanical or electri-
 165 cal stress and $\varphi_i(u) = \kappa_i u^{n_i}$, with $\kappa_i > 0$ a scale factor and $n_i > 0$ the stress
 166 exponent. Choosing h_i recovers the exponential, power-law and stretched-
 167 exponential degradation classes [3]; the results below use only $\varphi_i > 0$ and
 168 $h_i > 0$.

169 The within-unit dynamics are deterministic, where most of the degrada-
 170 tion literature places the randomness inside a single unit's path [7, 8, 10].
 171 We put it between units instead (Section 2.2). The object of study is the
 172 direction along which information arrives, which is fixed by the relative rates
 173 of the competing mechanisms: a within-unit diffusion would jitter that di-
 174 rection without altering its systematic motion, whereas dispersion in the rate
 175 parameters across a population is what makes remaining life uncertain and
 176 therefore worth estimating.

177 We measure damage on a scale normalised so that mechanism i exhausts
 178 the component's tolerance when x_i reaches unity. A component is withdrawn
 179 at a condemnation limit at or below that point, so we write the end of life
 180 as the first passage of any one mechanism to a threshold $x_f \in (0, 1]$:

$$T_f = \inf\{t \geq 0 : \max_i x_i(t) \geq x_f\}. \quad (3)$$

181 No result below depends on the value of x_f , which enters only by fixing where
 182 a trajectory stops. The computed quantities of Section 6 do depend on it,
 183 and the value used there is stated with the rest of the test plant.

184 2.2. What is to be estimated, and why it is uncertain

185 The quantity a prognostic system must report is the remaining useful life

$$R(t) := T_f - t, \quad (4)$$

186 the time left before failure.

187 We must say where the uncertainty in R lies, because (2) is a deterministic
 188 differential equation: for one component whose initial damage and parame-
 189 ters were known exactly, T_f and hence R would be known exactly, and there
 190 would be nothing to estimate. The uncertainty is between components, not
 191 within one. Units built to the same specification differ in initial damage and
 192 in the coefficients of (2), which is the unit-to-unit variability that reliability

193 engineering treats as the primary source of scatter in life data [22]. A prog-
 194 nostic algorithm observes one unit drawn from that population and must
 195 infer its remaining life.

196 We therefore treat R , at any given observation instant, as a random
 197 variable over the population of units, with prior variance $\text{Var}[R] > 0$. This
 198 variance is the uncertainty the algorithm exists to reduce, and it is the reason
 199 the estimation problem is not trivial. Everything below is written for a
 200 nominal trajectory of (2), with the fleet spread entering only through $\text{Var}[R]$.

201 2.3. Observations and the informative direction

202 Each damage state reaches the record through its own noisy channel,

$$y_i = x_i + \sigma_i \epsilon_i, \quad i = 1, \dots, d, \quad (5)$$

203 with $\sigma_i > 0$ the noise level of channel i and ϵ_i independent standard normal
 204 variables. The record available to the algorithm is the collection of these
 205 measurements over the observation window.

206 Equation (5) pairs each mechanism with a channel of its own. Condition
 207 monitoring in general returns $y = H(x, u) + \epsilon$, with sensors that mix mech-
 208 anisms, and the pairing is an assumption rather than a description. It is
 209 load-bearing: the positivity of $\mathbf{d}$ that Remark 3.1 shows the results to need
 210 follows from $\varphi_i > 0$ and $h_i > 0$ in these coordinates and need not survive
 211 an arbitrary H . Where H is known, invertible and positivity-preserving the
 212 construction carries over to $H^{-1}y$; the general case does not reduce to it.

213 Two assumptions carry the sensitivity, and both are stated here rather
 214 than left to be inferred.

215 *The prior on R .* We take R to be Gaussian over the population with variance
 216 $\text{Var}[R]$. Linearising the record about the nominal trajectory, following [3],
 217 then makes the record and R jointly Gaussian, and the mutual informations
 218 below are exact. Linearisation alone would not do it: with a non-Gaussian
 219 prior of the same variance the Gaussian value is an upper bound on what
 220 a channel carries, and Lemma 2.2 becomes an inequality rather than an
 221 identity. The information geometry that follows is exact under this Gaus-
 222 sian prior and the time-shift population described next; it is not a model of
 223 population variability in general.

224 *How units differ.* We take the population spread to be a reparametrisation
 225 of a single degradation path: units differ in where they sit on the trajectory
 226 of (2), not in its shape. This is what makes R a coordinate along that path,
 227 and it is what the next display uses. Dispersion in the coefficients of (2) is a
 228 different matter, because it moves $\partial x/\partial \theta$ rather than sliding along $\partial x/\partial R$,
 229 and Remark 2.1 records what it costs.

230 The sensitivity of channel i to remaining life then follows from (4): be-
 231 cause R falls at unit rate as t advances,

$$\frac{\partial x_i}{\partial R} = -\frac{\partial x_i}{\partial t} = -\varphi_i(u) h_i(x_i). \quad (6)$$

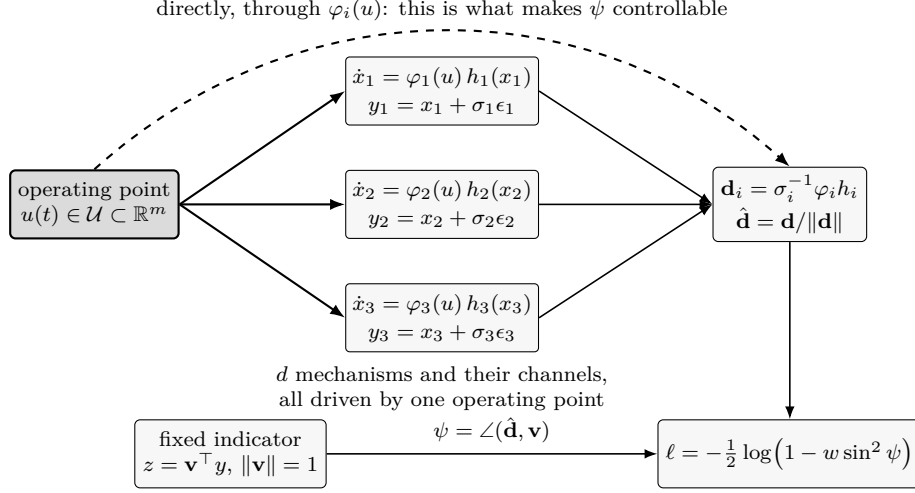
232 **Remark 2.1** (what the time-shift assumption costs). *Equation (6) is exact*
 233 *when the population spread slides along one path, and only then. On the*
 234 *plant of Section 6.1, with units instead drawn by dispersing the degradation*
 235 *coefficients, the direction that the population regression of x on R actually*
 236 *returns stands about 45° from $-\varphi(u)h(x)$, across two families of dispersed*
 237 *coefficient and every spread tried; a control whose units are started on the*
 238 *reference trajectory, where (6) holds by construction, returns the same re-*
 239 *gression to within 0.2° . The misalignments this paper is about are a few*
 240 *degrees. Extending the construction to populations dispersed in θ , where the*
 241 *observation carries scatter that no value of R explains, is open.*

242 Dividing each channel by its own noise level puts them on a common
 243 scale, and taking magnitudes gives the whitened signal direction

$$\mathbf{d}(t) = \left[\sigma_i^{-1} \varphi_i(u) h_i(x_i) \right]_{i=1}^d, \quad \hat{\mathbf{d}} = \mathbf{d}/\|\mathbf{d}\|. \quad (7)$$

244 In words: $\mathbf{d}$ says how strongly, and in what combination of channels, the
 245 measurement responds to a change in remaining life. Its direction $\hat{\mathbf{d}}$ is the
 246 *informative direction*, the combination a health indicator has to be aligned
 247 with and the object named informally in Section 1. Its length $\|\mathbf{d}\|$ is the
 248 strength of that signal. The two objects carry the paper: $\|\mathbf{d}\|$ is how much
 249 information arrives and $\hat{\mathbf{d}}$ is where it arrives from, and the loss of a fixed
 250 indicator depends only on the second. Experiment design in the observability
 251 tradition scores the first, which is why Proposition 4.6 finds it blind to the
 252 loss.

253 We use two properties of (7) repeatedly below.



Alignment asks $\hat{\mathbf{d}}(x, u) = q$: $d - 1$ scalar equations on m unknowns, whence the counting rule of Theorem 4.1.

Figure 1: The construction. The operating point enters twice, through the plant and directly through $\varphi_i(u)$; the second path is what makes the misalignment ψ a control variable. One input drives all d mechanisms.

254 *Positivity.* Every entry of $\mathbf{d}$ is strictly positive, because $\varphi_i > 0$ and $h_i >$
 255 0 in (2). This is a consequence of the degradation model rather than an
 256 assumption imposed for convenience, and Remark 3.1 shows that the results
 257 below would fail without it.

258 *Actuation.* $\mathbf{d}$ is an explicit algebraic function of the state and the input.
 259 The direction along which information about R arrives is therefore not a
 260 fixed property of the instrumentation: it moves when the operating point
 261 moves. That is the observation this paper is built on, and Fig. 1 places it:
 262 the operating point reaches $\hat{\mathbf{d}}$ twice, once through the state and once directly.

263 Because R is scalar, a single signal-to-noise ratio governs the information
 264 the record carries about it, namely the squared sensitivity times the prior
 265 uncertainty,

$$\gamma = \|\mathbf{d}\|^2 \text{Var}[R]. \quad (8)$$

266 Rates in (7) are taken with respect to clock time, whereas [3] differentiates
 267 with respect to age normalised by T_f . The two conventions differ by a factor
 268 common to all d channels, so they give the same $\hat{\mathbf{d}}$, and hence the same

269 geometry in Sections 3–4, while rescaling γ . Section 5, where lifetime enters
 270 the objective, restores the normalised convention.

271 2.4. The fixed health indicator and its information loss

272 A health indicator compresses the d measured channels into one scalar
 273 before any life prediction is made. We take it to be a fixed linear map.
 274 Since (7) already measures each channel in units of its own noise, we write
 275 the indicator in those same coordinates: with $\Sigma := \text{diag}(\sigma_1^2, \dots, \sigma_d^2)$ and
 276 $\tilde{y} := \Sigma^{-1/2}y$ the whitened record,

$$z = \mathbf{v}^\top \tilde{y}, \quad \|\mathbf{v}\| = 1, \quad (9)$$

277 with the weight vector $\mathbf{v}$ chosen once, at design time, and held for the whole
 278 life of the component. Fixing it is not a modelling convenience: the map
 279 is typically embedded in firmware or frozen by certification, so it cannot be
 280 re-tuned as the component ages. An indicator specified on the raw measure-
 281 ments, $z = \mathbf{w}^\top y$, is the same map under $\mathbf{v} \propto \Sigma^{1/2}\mathbf{w}$, and the normalisation
 282 $\|\mathbf{v}\| = 1$ only fixes the scale of z ; it restricts nothing. The distinction matters
 283 because the angle below is measured after whitening: unless the σ_i are equal,
 284 a unit vector in raw coordinates is not a unit vector in whitened ones, and
 285 the two readings of ψ disagree.

286 We compute the cost of fixing $\mathbf{v}$ as follows. Write $\psi(t)$ for the angle
 287 between the informative direction $\hat{\mathbf{d}}(t)$ and the indicator weights $\mathbf{v}$. Af-
 288 ter whitening, the signal in the record points along $\hat{\mathbf{d}}$ and has magnitude
 289 $\|\mathbf{d}\| \sqrt{\text{Var}[R]}$, while the noise is isotropic of unit scale. Projecting onto a unit
 290 vector $\mathbf{v}$ therefore keeps the fraction $\cos \psi$ of the signal and all of the noise,
 291 so the indicator’s signal-to-noise ratio is

$$\gamma \cos^2 \psi \quad (10)$$

292 against the γ of (8) available to an oracle that is free to re-aim its projection at
 293 $\hat{\mathbf{d}}(t)$ at every instant. Since a scalar Gaussian channel of signal-to-noise ratio
 294 s carries $\frac{1}{2} \log(1 + s)$ nats about R [23], the information the fixed indicator
 295 forfeits, at each instant, is

$$\ell = \frac{1}{2} \left[\log(1 + \gamma) - \log(1 + \gamma \cos^2 \psi) \right] \geq 0, \quad (11)$$

296 The design objective throughout is its average over the life of the component,

$$\bar{\ell} := \frac{1}{T_f} \int_0^{T_f} \ell(t) dt, \quad (12)$$

297 a loss per unit time. We optimise this rate rather than the information the
 298 whole observation record forfeits, because (12) is unaffected by how densely
 299 the record is sampled and is therefore a property of the indicator rather
 300 than of the measurement schedule. The two are different quantities, and
 301 Proposition 2.3 gives the second in closed form.

302 A third quantity is the one an operator actually pays for: the error of
 303 whatever remaining-life estimator runs downstream. It is neither of the first
 304 two and is not a function of them. The loss is a logarithmic information
 305 quantity and the error is a time, and an estimator holding a whole history
 306 recovers much of what any single instant discards, so a large spread in (12)
 307 does not imply a comparable spread in error. What the theory can offer
 308 downstream is therefore an ordering rather than a conversion, and Section 6.8
 309 tests that ordering against an estimator built from none of this construction:
 310 it survives, with a spread in error much smaller than the spread in loss.
 311 Equation (12) should be read accordingly, as an information-geometric ob-
 312 jective whose ranking carries downstream, not as a surrogate for prognostic
 313 accuracy.

314 Two remarks on what (11) does and does not say. It is a loss relative to
 315 an oracle that no fixed map can realise, so it measures the price of the time-
 316 invariance constraint rather than of a bad design; a perfectly chosen $\mathbf{v}$ still
 317 pays it whenever $\hat{\mathbf{d}}$ moves. And it depends on the indicator only through the
 318 single angle ψ : two indicators that make the same angle with $\hat{\mathbf{d}}$ are equally
 319 good, whatever else distinguishes them.

320 The next lemma replaces (11) by a closed form that will be more conven-
 321 nient throughout, and records how it behaves at small misalignment.

322 **Lemma 2.2.** *Let $w := \gamma/(1 + \gamma) \in (0, 1)$. Then*

$$\ell = -\frac{1}{2} \log(1 - w \sin^2 \psi), \quad (13)$$

323 *exactly and at every signal-to-noise ratio, and*

$$\ell = \frac{w}{2} \psi^2 + \frac{w(3w-2)}{12} \psi^4 + O(\psi^6). \quad (14)$$

324 *Proof.* Substituting $\cos^2 \psi = 1 - \sin^2 \psi$ into (11) gives

$$\ell = -\frac{1}{2} \log \frac{1 + \gamma - \gamma \sin^2 \psi}{1 + \gamma} = -\frac{1}{2} \log(1 - w \sin^2 \psi).$$

325 For (14), use $-\frac{1}{2} \log(1-s) = \frac{s}{2} + \frac{s^2}{4} + O(s^3)$ with $s = w \sin^2 \psi$ and $\sin^2 \psi = \psi^2 -$
 326 $\psi^4/3 + O(\psi^6)$; collecting the ψ^4 terms gives $-w/6 + w^2/4 = w(3w-2)/12$. $\square$

327 The quadratic term of (14) is the bound obtained in [3, Thm. 1], which is
 328 therefore the leading term of an exact identity rather than an approximation
 329 of unknown quality. The quartic coefficient $w(3w - 2)/12$ changes sign at
 330 $w = 2/3$, that is at $\gamma = 2$: below that signal-to-noise ratio the quadratic
 331 model overstates the loss, above it the model understates it. This matters in
 332 practice because prognostic problems sit on both sides of $\gamma = 2$: an early-life
 333 measurement with a weak signal on one side, a well-instrumented component
 334 near failure on the other. Fig. 2(a) plots the exact loss against its quadratic
 335 approximation on both sides of that ratio.

336 **Proposition 2.3.** *Let the record consist of measurements (5) at instants*
 337 *$t_1, \dots, t_N$, with the noise independent across channels and across instants,*
 338 *and write $\mathbf{d}_k := \mathbf{d}(t_k)$ and ψ_k for the angle between $\hat{\mathbf{d}}_k$ and $\mathbf{v}$. Put*

$$G := \text{Var}[R] \sum_{k=1}^N \|\mathbf{d}_k\|^2, \quad W := \frac{G}{1 + G},$$

$$S := \frac{\sum_k \|\mathbf{d}_k\|^2 \sin^2 \psi_k}{\sum_k \|\mathbf{d}_k\|^2}.$$

339 *Then the information about R that the fixed indicator forfeits over the whole*
 340 *record is*

$$\ell_{\text{rec}} = -\frac{1}{2} \log(1 - WS). \quad (15)$$

341 *Proof.* See Supplementary Section S2. □

342 Equation (15) is (13) again, with the instantaneous signal-to-noise ratio
 343 γ replaced by its sum G over the record, and $\sin^2 \psi$ by the mean S weighted
 344 by the signal energy $\|\mathbf{d}_k\|^2$. The reduction to one angle therefore survives the
 345 passage from an instant to a record. What does not survive is the averaging:
 346 $\bar{\ell}$ averages the loss, whereas (15) averages $\sin^2 \psi$ inside it, and the two differ.
 347 They also select different indicators. Minimising (15) over $\|\mathbf{v}\| = 1$ maximises
 348 $\mathbf{v}^\top \left(\sum_k \mathbf{d}_k \mathbf{d}_k^\top \right) \mathbf{v}$, so the best fixed indicator for the record is the leading
 349 eigenvector of the integrated Fisher matrix, while the minimiser of $\bar{\ell}$ is the
 350 barycentre of Proposition 3.3. Every result below is stated for $\bar{\ell}$.

351 **Corollary 2.4** (what the loss costs an estimator). *In the model of Section 2,*
 352 *the minimum-variance estimate of R formed from the fixed indicator carries*
 353 *mean-squared error*

$$\frac{\text{Var}[R \mid z]}{\text{Var}[R \mid \hat{y}]} = \frac{1 + \gamma}{1 + \gamma \cos^2 \psi} = e^{2\ell}$$

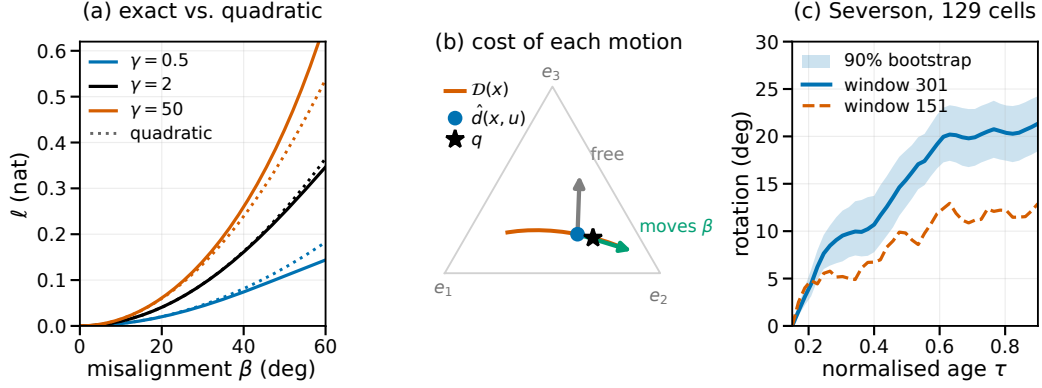


Figure 2: (a) Lemma 2.2: the exact loss (solid) against its quadratic model (dotted), whose quartic coefficient changes sign at $\gamma = 2$. (b) Theorem 3.6: of the replicator velocity, only the component along the geodesic towards q moves β . (c) The rotation, measured on the Severson fleet at two differentiation windows.

354 times that of the estimate formed from the whole whitened record, at each
 355 instant. The same identity holds over a record with ℓ replaced by ℓ_{rec} of
 356 Proposition 2.3.

357 *Proof.* Write $\tilde{y} = \mathbf{d}R + \epsilon$ with $\epsilon \sim N(0, I_d)$ and R Gaussian. The indicator
 358 returns $z = (\mathbf{v}^\top \mathbf{d})R + \mathbf{v}^\top \epsilon$, whose noise is standard normal because $\|\mathbf{v}\| = 1$,
 359 so its signal-to-noise ratio is $\gamma \cos^2 \psi$ and $\text{Var}[R | z] = \text{Var}[R]/(1 + \gamma \cos^2 \psi)$.
 360 The record's sufficient statistic is $\mathbf{d}^\top \tilde{y}$, with ratio γ , giving $\text{Var}[R | \tilde{y}] =$
 361 $\text{Var}[R]/(1 + \gamma)$. Their quotient is $e^{2\ell}$ by (11). The record statement is the
 362 same computation with G and S in place of γ and $\sin^2 \psi$. $\square$

363 The loss is therefore not a score standing in for estimation error: it is the
 364 logarithm of the factor by which that error inflates. What the corollary does
 365 not settle is whether the ordering survives an estimator other than the one
 366 the model prescribes, and Section 6.8 checks that separately.

367 3. The information simplex and its replicator flow

368 Lemma 2.2 reduced the loss to one angle. This section shows that the
 369 angle lives on a familiar object, the probability simplex, and that the degra-
 370 dation traces a flow along it that the operating point drives. The payoff
 371 is threefold: the loss becomes a recognised statistical distance, the optimal
 372 indicator becomes a barycentre, and the effect of the input becomes a single

interpretable quantity. Two of the results here are load-bearing for what follows — the replicator form of Theorem 3.4 and the misalignment rate of Theorem 3.6, which Section 4 builds the reachability argument on. The Bhattacharyya identity and the barycentre are interpretation: they say what the loss and the optimal indicator already are in recognised terms, and nothing later depends on them.

3.1. Information shares

Because every entry of $\mathbf{d}$ is positive (Section 2.3), the squared components of the unit vector $\hat{\mathbf{d}}$ form a probability vector:

$$p_i := \hat{\mathbf{d}}_i^2, \quad p_i > 0, \quad \sum_i p_i = 1. \quad (16)$$

The interpretation is direct. The total Fisher information the record carries about R is $\|\mathbf{d}\|^2 \text{Var}[R]$ by (8), and channel i contributes $\mathbf{d}_i^2 \text{Var}[R]$ of it. So p_i is exactly the share of the total information about remaining life that arrives through channel i . A component in which one mechanism dominates the degradation has p near a vertex of the simplex; one in which three mechanisms contribute equally sits near its centre.

The same map applied to the indicator, $q_i := \mathbf{v}_i^2$, is legitimate because an optimal indicator may be taken with non-negative weights.

Remark 3.1 (positivity is load-bearing). *If $\hat{\mathbf{d}} \geq 0$ then $\hat{\mathbf{d}}^\top |\mathbf{v}| \geq |\hat{\mathbf{d}}^\top \mathbf{v}|$ at every instant, so replacing $\mathbf{v}$ by $|\mathbf{v}|$ never increases (11): non-negative weights are without loss of generality. This argument uses the positivity of $\mathbf{d}$ inherited from $\varphi_i, h_i > 0$, and it fails without it: in random trials with sign-indefinite $\hat{\mathbf{d}}$, taking absolute values was worse than the signed vector about a third of the time. The results therefore require the positive-orthant structure of the degradation model; it is not only a notational convenience.*

3.2. The loss as a Bhattacharyya distance

The identification below is used twice in what follows: it gives “the average of a curve on the simplex” an unambiguous meaning (Section 3.3), and it lets the high-signal limit of the loss be read off as a known divergence.

Proposition 3.2. *$\cos \psi = \sum_i \sqrt{p_i q_i}$ is the Bhattacharyya coefficient of p and q . Consequently $\psi = \frac{1}{2} d_{\text{FR}}(p, q)$, where d_{FR} is the Fisher–Rao distance on the simplex, and, writing $D_{\text{B}}(p, q) = -\log \sum_i \sqrt{p_i q_i}$ for the Bhattacharyya divergence,*

$$\ell = -\frac{1}{2} \log \left[(1 - w) + w e^{-2D_{\text{B}}(p, q)} \right]. \quad (17)$$

405 As $\gamma \rightarrow \infty$, $\ell \rightarrow D_{\text{B}}(p, q)$.

406 *Proof.* By construction,

$$\cos \psi = \hat{\mathbf{d}}^\top \mathbf{v} = \sum_i \sqrt{p_i} \sqrt{q_i},$$

407 which is precisely the Bhattacharyya coefficient of p and q . The map $p \mapsto \sqrt{p}$
 408 is the classical isometry between the simplex under the Fisher–Rao metric
 409 and the positive orthant of the unit sphere, under which the geodesic distance
 410 is twice the spherical angle [18, 19]; hence $\psi = \frac{1}{2}d_{\text{FR}}$. Then $\cos^2 \psi = e^{-2D_{\text{B}}}$,
 411 and substituting into (13) gives (17). Letting $w \rightarrow 1$ gives the limit. $\square$

412 The square root is a bijection between the simplex and the non-negative
 413 part of the unit sphere, carrying p to $\hat{\mathbf{d}}$ and q to $\mathbf{v}$, and by the proposition
 414 it is an isometry. We use the two descriptions interchangeably from here
 415 on: a statement such as $q \in \mathcal{D}(x)$ in Section 4 says that $\mathbf{v}$ is a reachable
 416 direction, and a distance between points of the simplex is the angle between
 417 the corresponding unit vectors.

418 The resulting picture is the one the rest of the paper develops. The
 419 degrading component is a curve on the simplex, the health indicator is a
 420 single point on it, and the information lost depends only on the distance
 421 between them. Design of the indicator is the choice of one point; control of
 422 the operating point is the shaping of the curve.

423 3.3. The optimal indicator as a barycentre

424 If the indicator is one point and the plant is a curve, the best point should
 425 be some kind of average of the curve. It is, but of two different kinds in two
 426 different limits.

427 **Proposition 3.3.** *Let $q^\star$ minimise the time-averaged loss.*

- 428 (i) *As $\gamma \rightarrow \infty$, $\ell \rightarrow D_{\text{B}}$ pointwise, so $q^\star$ tends to the Bhattacharyya*
 429 *barycentre $\arg \min_q \int D_{\text{B}}(p_\tau, q) d\tau$.*
- 430 (ii) *As the curve shortens, $\ell = \frac{w}{2}\psi^2 + O(\psi^4)$, so $q^\star$ tends to the w -weighted*
 431 *Fisher–Rao barycentre, and $\bar{\ell} = \frac{1}{8}\bar{w} \text{Var}^{\text{FR}} + O(\psi^4)$ where $\text{Var}^{\text{FR}} :=$*
 432 *$\min_q \mathbb{E}_\mu[d_{\text{FR}}^2(p_\tau, q)]$ is the μ -weighted Fréchet variance of the curve.*

433 *Proof.* See Supplementary Section S2. $\square$

434 The two limits are taken in different parameters, signal-to-noise ratio in
 435 (i) and arc length in (ii), and they do not coincide. On a curve of angular
 436 span 82.7° the two barycentres lie 0.72° apart, and q^* crosses from the Fisher–
 437 Rao side to the Bhattacharyya side between 0 and 10 dB, reaching 3×10^{-5}
 438 degrees of the latter by 50 dB. For design purposes the practical consequence
 439 is mild: either barycentre of a nominal degradation path is a good starting
 440 indicator.

441 3.4. The replicator flow and the role of the input

442 We now ask how fast the informative direction moves and what the op-
 443 erating point does to that motion.

444 **Theorem 3.4.** *Let $g_i = \log \mathbf{d}_i$ denote the logarithmic rate of channel i .
 445 Along (2),*

$$\dot{\mathbf{d}}_i = \hat{\mathbf{d}}_i(\dot{g}_i - \mathbb{E}_p[\dot{g}]), \quad (18)$$

446 *so the information shares obey the replicator equation of evolutionary dynam-*
 447 *ics, in which each share grows in proportion to the amount by which its own*
 448 *rate exceeds the average, the average being taken over the shares themselves:*

$$\dot{p}_i = 2p_i(\dot{g}_i - \mathbb{E}_p[\dot{g}]), \quad (19)$$

449 *and*

$$\|\dot{\mathbf{d}}\| = \text{Std}_p[\dot{g}], \quad \dot{g} = \left[\varphi_i(u) h'_i(x_i) \right]_i + \nabla_u \log \varphi \cdot \dot{u}. \quad (20)$$

450 *Proof.* From $\mathbf{d}_i = e^{g_i}$ we have $\dot{\mathbf{d}}_i = \mathbf{d}_i \dot{g}_i$, so $\dot{\mathbf{d}} = \|\mathbf{d}\|^{-1}(\dot{\mathbf{d}} - \hat{\mathbf{d}}(\hat{\mathbf{d}}^\top \dot{\mathbf{d}}))$ has
 451 components $\hat{\mathbf{d}}_i \dot{g}_i - \hat{\mathbf{d}}_i \sum_j \hat{\mathbf{d}}_j^2 \dot{g}_j$, which is (18); squaring and summing gives
 452 $\text{Var}_p[\dot{g}]$. Since $p_i = \hat{\mathbf{d}}_i^2$, $\dot{p}_i = 2\hat{\mathbf{d}}_i \dot{\mathbf{d}}_i = 2p_i(\dot{g}_i - \mathbb{E}_p[\dot{g}])$. Differentiating $g_i =$
 453 $\log \varphi_i(u) + \log h_i(x_i) - \log \sigma_i$ along (2) gives $\dot{g}_i = \nabla_u \log \varphi_i \cdot \dot{u} + (h'_i/h_i)\varphi_i h_i$,
 454 and $(h'_i/h_i)\varphi_i h_i = \varphi_i h'_i$. $\square$

455 Equation (20) has a compact reading. The speed at which the informative
 456 direction rotates equals the information-weighted spread of the mechanisms'
 457 logarithmic rates. If all mechanisms accelerate at the same proportional rate
 458 the direction does not move at all, however fast the component is degrading;
 459 it is disagreement among the mechanisms that rotates it. The identity is the
 460 Fisher–Rao/Shahshahani geometry of the simplex combined with Fisher's
 461 fundamental theorem of natural selection [18, 20, 21]. What is new here is

its identification with the informative direction of a degrading component, and, decisively, that the quantity playing the role of fitness, the logarithmic rate $\dot{g}_i$ of channel i , is actuated: the second summand in (20) is the operator's lever.

Remark 3.5 (contrast with Oja's flow). *Oja's flow for online principal-component extraction has the same algebraic form; for a diagonal covariance it reduces to a replicator equation with fitness twice the eigenvalues [24]. The roles are opposite. There the flow is an algorithm, run to fixation in order to find a direction, with constant fitness. Here it is the plant, its fitness is time-varying and actuated, and the aim is to prevent fixation from carrying the direction away from a fixed indicator.*

Only one scalar matters for the loss. Fix the indicator q and write $\beta := \angle(\hat{\mathbf{d}}, \mathbf{v}) = \frac{1}{2}d_{\text{FR}}(p, q)$ for the misalignment. This is the angle ψ of Lemma 2.2 under a new name, and the change of name marks a change of role: ψ was a fixed angle between two directions, whereas $\beta = \beta(x_t, u_t)$ is followed along the trajectory and is the quantity the input acts on. Every loss expression below may therefore be read as a function of β .

Theorem 3.6. *Let the operating-point trajectory $u(\cdot)$ be absolutely continuous, and let $\hat{e}$ be the unit tangent at $\hat{\mathbf{d}}$ along the geodesic towards $\mathbf{v}$. Then $\dot{\beta} = -\langle \dot{\hat{\mathbf{d}}}, \hat{e} \rangle$, which is affine in the input rate: $\dot{\beta} = \alpha_0(x, u) + \alpha(x, u)^\top \dot{u}$. In particular $|\dot{\beta}| \leq \|\dot{\hat{\mathbf{d}}}\|$, with equality only for motion along that geodesic.*

Proof. $\cos \beta = \hat{\mathbf{d}}^\top \mathbf{v}$ gives $-\sin \beta \dot{\beta} = \dot{\hat{\mathbf{d}}}^\top \mathbf{v}$. Also $\hat{e} = (\mathbf{v} - \cos \beta \hat{\mathbf{d}}) / \sin \beta$ and $\dot{\hat{\mathbf{d}}}^\top \hat{\mathbf{d}} = 0$, so $\langle \dot{\hat{\mathbf{d}}}, \hat{e} \rangle = \dot{\hat{\mathbf{d}}}^\top \mathbf{v} / \sin \beta = -\dot{\beta}$. Affinity follows because $\dot{g}$ is affine in $\dot{u}$ by (20) and $\dot{\hat{\mathbf{d}}}$ is linear in $\dot{g}$. $\square$

The hypothesis is needed only to write $\dot{u}$; nothing in Section 4 depends on it, since the reachable sets (21) are defined pointwise in u and the limits below hold for every measurable admissible input. Theorem 3.6 is a local kinematic description of smooth steering, not a restriction on the controls the rest of the paper admits.

The component of the replicator velocity orthogonal to $\hat{e}$ is therefore free: the informative direction may rotate in a $(d-2)$ -dimensional subspace at no cost in information. Fig. 2(b) shows the decomposition at one state. This is the observation that separates the two objectives of the next section.

495 4. Reachability limits on the misalignment

496 Controlling the rate $\dot{\beta}$ is not the same as reaching a value of β . The
 497 distinction is the subject of this section, and it is what decides whether a
 498 fixed indicator can be made lossless at all.

499 For a state x , the set of informative directions the operator can produce
 500 by choosing the operating point is

$$\mathcal{D}(x) := \{\hat{\mathbf{d}}(x, u) : u \in \mathcal{U}\} \subset S^{d-1}, \quad (21)$$

501 the reachable direction set, and the closest the indicator can be brought at
 502 that state is the reachable misalignment floor

$$\beta_{\min}(x, q) := \text{dist}(q, \mathcal{D}(x)). \quad (22)$$

503 For a single input, $\mathcal{D}(x)$ is an arc traced out as u sweeps $\mathcal{U}$; its length measures
 504 the actuation authority available at x , and it shrinks to a point when the
 505 mechanisms respond identically to the input.

506 **Theorem 4.1.** *Fix an admissible input and let $x(\cdot)$ be the trajectory it gen-*
 507 *erates. Zero loss along that trajectory is attainable with the fixed indicator q*
 508 *if and only if $q \in \bigcap_t \mathcal{D}(x_t)$. If $m < d - 1$ and $u \mapsto \hat{\mathbf{d}}(x, u)$ is continuously*
 509 *differentiable, the indicators that attain zero loss form a null subset of the*
 510 *unit sphere, so almost every q , and in fact every q that does not attain it, has*
 511 *$\beta_{\min} > 0$ on a set of times of positive measure. Whether any q at all attains*
 512 *zero loss is a property of the plant, and Remark 4.5 exhibits plants for which*
 513 *one does. In every case $\bar{\ell} \geq \int \ell(\beta_{\min}(x_t, q)) dt/T_f$ for that trajectory.*

514 *Proof.* By Lemma 2.2, $\ell(\beta) = 0$ iff $\sin \beta = 0$, so $\bar{\ell} = 0$ iff $\beta_t = 0$ for almost
 515 every t , that is iff $\hat{\mathbf{d}}(x_t, u_t) = q$, that is iff $q \in \mathcal{D}(x_t)$ for every t . Solving
 516 $\hat{\mathbf{d}}(x, u) = q$ imposes $d - 1$ independent scalar equations, both sides being
 517 unit vectors, on m unknowns. If $m < d - 1$ then $\mathcal{D}(x)$, the image of an
 518 m -dimensional set under a continuously differentiable map, is a null subset
 519 of the $(d - 1)$ -dimensional sphere, which almost every q misses; if $m \geq d - 1$
 520 the implicit function theorem supplies a solution wherever the Jacobian has
 521 full rank. The inequality $\beta_t \geq \beta_{\min}(x_t, q)$ holds for every admissible input by
 522 definition of $\beta_{\min}$ as a minimum over $\mathcal{U}$, and is strict exactly where $q \notin \mathcal{D}(x_t)$.
 523 Since $\mathcal{U}$ is compact and $\hat{\mathbf{d}}$ continuous, each $\mathcal{D}(x_t)$ is compact and varies
 524 continuously with t , so the set of times at which $q \notin \mathcal{D}(x_t)$ is open. It is

525 non-empty for every q that does not attain zero loss, in particular for every q
 526 outside the null set $\mathcal{D}(x_0)$, and so has positive measure. Since ℓ is increasing
 527 on $[0, \pi/2]$, the bound follows on dividing by T_f . $\square$

528 This is a state-conditioned result. The bound is genuine, in the sense
 529 that no input violates a reachability constraint at the state it occupies, but
 530 it is conditional on the trajectory, which the input also steers. Optimising
 531 over the trajectory as well is deferred to Section 5, and the two levels are the
 532 structure of the argument: a state-conditioned limit here, a trajectory-level
 533 limit there.

534 The practical reading of Theorem 4.1 is a counting rule: a component
 535 degrading through d mechanisms generically needs $d - 1$ independently ad-
 536 justable operating variables before a fixed indicator can be kept perfectly
 537 aligned. Here *generically* is a count of dimensions over the plant's data,
 538 not a statement of measure: the theorem proves the measure statement over
 539 indicators, Remark 4.5 constructs plants that escape the count, and we do
 540 not prove those plants rare. Section 4.1 says what the qualifier covers. For a
 541 generic plant under this model, three mechanisms and a single temperature
 542 setpoint will not do it. The three results of this section answer different ques-
 543 tions, and it is worth separating them. Theorem 4.1 characterises zero loss
 544 exactly, as membership of every reachable set, and then counts dimensions to
 545 say when a generic plant can meet it; the count is necessary for generic data
 546 and is not by itself a construction. Proposition 4.4 gives an exact criterion,
 547 necessary and sufficient, on the separable subclass alone. Remark 4.5 exhibits
 548 plants outside that subclass which attain zero loss with fewer inputs than the
 549 count asks, so the count is not necessary in general. But a weaker and much
 550 cheaper objective is available.

551 **Corollary 4.2** (what actuation is actually required). *Two objectives that are*
 552 *easily conflated carry different requirements.*

- 553 (i) *Driving $\bar{\ell}$ to zero is equivalent to freezing $\hat{\mathbf{d}}$ at q , and needs $d - 1$*
 554 *conditions on $\dot{u}$, hence generically $m \geq d - 1$.*
- 555 (ii) *Holding β at a constant level, not necessarily zero, is the single scalar*
 556 *equation $\alpha_0 + \alpha^\top \dot{u} = 0$, solvable whenever $\alpha \neq 0$ and the resulting $u(\cdot)$*
 557 *stays in $\mathcal{U}$; generically $m \geq 1$ suffices.*

558 *Proof.* (i) By Theorem 4.1, $\bar{\ell} = 0$ iff $\hat{\mathbf{d}}(x_t, u_t) = q$ for every t , so $\hat{\mathbf{d}}$ is constant;
 559 by Theorem 3.4 that forces $\dot{g}_i = \mathbb{E}_p[\dot{g}]$ for every i , which is $d - 1$ independent

560 conditions on $\dot{u} \in \mathbb{R}^m$. (ii) is one equation in $\dot{u}$ by Theorem 3.6. The gap
 561 between them is the content of that theorem: $\dot{\beta}$ sees only the component of
 562 $\hat{\mathbf{d}}$ along $\hat{e}$, so the remaining $d - 2$ directions of rotation are unconstrained by
 563 (ii) and constrained by (i). $\square$

564 Minimising $\bar{\ell}$ is neither of these: it trades the level of β against the
 565 trajectory that carries it, and is the problem of Section 5.

566 4.1. The reachable sets are translates of one set

567 Theorem 4.1 counts equations against unknowns and hedges with
 568 “generic”. The hedge is not idle, and the following observation says exactly
 569 what it covers.

570 **Lemma 4.3** (separability). *Write $\xi(x, u)$ for the class of $\log \mathbf{d}(x, u)$ modulo*
 571 *the all-ones vector, which is what $\hat{\mathbf{d}}$ retains. Then*

$$\begin{aligned} \xi(x, u) &= a(x) + b(u), \\ a_i(x) &= \log \frac{h_i(x_i)}{\sigma_i}, \quad b_i(u) = \log \varphi_i(u), \end{aligned} \tag{23}$$

572 *both taken modulo the all-ones vector. Consequently every reachable direction*
 573 *set is the same set translated: $\mathcal{D}(x) = a(x) + \mathcal{B}$ with $\mathcal{B} := b(\mathcal{U})$ fixed.*

574 *Proof.* Immediate from $\mathbf{d}_i = \sigma_i^{-1} \varphi_i(u) h_i(x_i)$ in (7): the logarithm turns the
 575 product into a sum, and the normalisation of $\hat{\mathbf{d}}$ removes any component along
 576 the all-ones vector. $\square$

577 The decomposition separates the two parties. The set $\mathcal{B}$ is fixed by the
 578 stress model φ and the envelope $\mathcal{U}$ alone (Fig. 3(a)). An operator’s authority
 579 can therefore be computed offline from the accelerated-life model, with no
 580 degradation record at all. The curve $\mathcal{A} := \{a(x_t)\}$ is fixed by the damage
 581 shapes h and the noise levels σ , and every reachable set is that one set carried
 582 along it (Fig. 3(b)). What the geometry of Section 3 measures is how far the
 583 plant’s demand $\mathcal{A}$ sticks out of the operator’s authority $\mathcal{B}$.

584 That offline route is the most useful consequence drawn in this paper
 585 and also the most restrictive. It holds on the separable subclass and not
 586 beyond it, and the best high-fidelity turbofan data we have reject separability
 587 (Section 7). This subsection and Proposition 4.4 should therefore be read as
 588 describing a structured subclass that is convenient when it applies, not as
 589 a general account of real machinery. Fig. 6 marks the one branch of the
 590 development that depends on it.

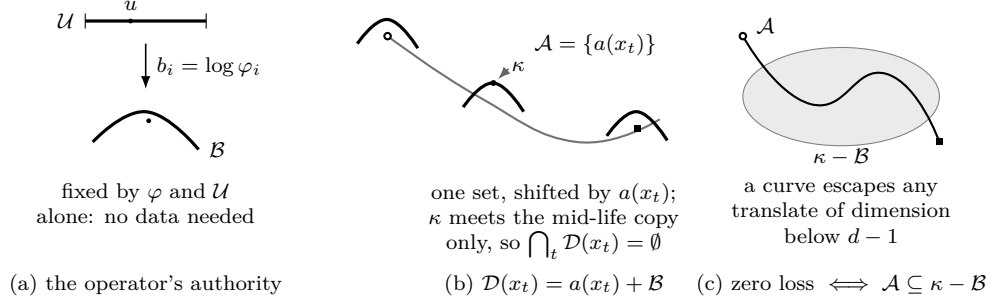


Figure 3: How the counting rule becomes a containment. (a) The operator's authority $\mathcal{B} = b(\mathcal{U})$, fixed by the stress model and the envelope alone. (b) Every reachable set is that one set translated, $\mathcal{D}(x_t) = a(x_t) + \mathcal{B}$ (circle: start of life, square: failure). (c) Zero loss asks the whole curve to lie in one translate (Proposition 4.4).

591 **Proposition 4.4** (exact criterion). *Zero loss is attainable along a trajectory*
 592 *if and only if $\mathcal{A}$ is contained in a translate of $-\mathcal{B}$. In particular $\mathcal{A}$ must lie*
 593 *in an affine subspace of dimension at most $\dim \mathcal{B} \leq m$, which for a curve*
 594 *in the $(d-1)$ -dimensional quotient fails generically unless $m \geq d-1$; this*
 595 *recovers Theorem 4.1. When it does not fail, $m < d-1$ suffices.*

596 *Proof.* By Theorem 4.1, zero loss holds iff $q \in \mathcal{D}(x_t)$ for every t . Writing κ
 597 for the quotient class of $\log \mathbf{v}$ and using Lemma 4.3, that reads $\kappa - a(x_t) \in \mathcal{B}$
 598 for every t , that is $\mathcal{A} \subseteq \kappa - \mathcal{B}$. The dimension statement follows because a
 599 translate of $\mathcal{B}$ spans an affine subspace of dimension $\dim \mathcal{B}$. $\square$

600 The containment is drawn in Fig. 3(c), and the mechanism the count
 601 obscures is now visible. Alignment at an instant fixes u_t outright, since
 602 $b(u_t) = \kappa - a(x_t)$ determines it; the input is spent, and any residual constraint
 603 on the state must then hold along the flow with no freedom left to enforce
 604 it. One input can carry the whole burden only when no residual constraint
 605 is left, which happens when $\mathcal{A}$ already lies parallel to $\mathcal{B}$.

606 **Remark 4.5** (a family with $m = 1$ and any d). *Take exponential shapes*
 607 *$h_i(x) = e^{\nu_i(x_i - x_i^0)}$, which (2) admits, and Arrhenius factors. Under the ramp*
 608 *$u(t) = u_0 + rt$ the damage rates are constant, $x_i(t) = x_i^0 + \varphi_i^0 t$ with $\varphi_i^0 =$*
 609 *$\varphi_i(u_0)$, so $a(x_t)$ moves at constant velocity $\nu_i \varphi_i^0$. Choosing $\nu_i = rE_i/\varphi_i^0$ makes*
 610 *that velocity parallel to E , which is the direction of $\mathcal{B}$, and $b(u_t)$ retreats*
 611 *along $\mathcal{B}$ at the same rate. The two cancel and $\hat{\mathbf{d}}$ is frozen. Supplementary*
 612 *Section S4 integrates the construction for $d = 2$ to 8.*

613 This bears on the sharpness of Theorem 4.1. The family of Remark 4.5
614 is a knife edge. At one percent of error in its shape coefficients, steering
615 is already worse than not steering (Supplementary Section S4). A designer
616 should read Proposition 4.4 as an explanation of when the counting rule
617 binds, and Theorem 4.1 as the rule to design against.

618 4.2. Invisibility to Fisher-information criteria

619 A natural objection is that all of this should follow from designing the in-
620 put to maximise the Fisher information about R , as in the observability-aware
621 literature discussed in Section 1. It does not, and the reason is structural.

622 **Proposition 4.6** (Fisher blindness). *The loss of a fixed indicator is not a*
623 *function of the Fisher information about R : configurations with equal Fisher*
624 *information can have different losses, and configurations with equal losses*
625 *can have different Fisher information. No criterion depending on the Fisher*
626 *information alone can therefore rank fixed indicators by the loss they incur.*

627 *Proof.* For a scalar latent the Fisher information is the number $\|\mathbf{d}\|^2 \text{Var}[R]$,
628 so trace, determinant and least eigenvalue coincide and every scalarisation is
629 a strictly monotone function of it. Under the rescaling $\mathbf{d} \mapsto c\mathbf{d}$ with $c > 0$,
630 that number changes while $\hat{\mathbf{d}}$, and hence β and ℓ , do not. A common gain
631 therefore equalises the scores of two configurations with different losses, and
632 separates the scores of two configurations with the same loss. $\square$

633 Stated in engineering terms, a Fisher criterion measures the strength of
634 the signal, whereas the loss of a fixed indicator depends on its direction.
635 Amplifying every channel by the same factor increases the first and leaves
636 the second unchanged. Section 6 exhibits this concretely: with the Fisher
637 score held equal to seven digits, the loss still spans a factor of 9.4.

638 5. The best achievable loss as a fractional control problem

639 Theorem 4.1 bounds the loss along a given trajectory. But the input
640 steers the trajectory as well as the misalignment on it, so the best achievable
641 loss requires optimising over both at once. This changes the structure of the
642 problem.

643 The objective is a time average over a lifetime that the input itself deter-
644 mines:

$$\bar{\ell}^*(q) = \min_{u(\cdot) \in \mathcal{U}} \frac{\int_0^{T_f} \ell(\beta(x_t, u_t)) dt}{T_f}. \quad (24)$$

Both the integrand and the upper limit T_f depend on $u(\cdot)$. This makes (24) a fractional optimal control problem, meaning that the objective is a ratio rather than that the dynamics carry a fractional-order derivative. The fraction is not a technicality: it is the source of the trade-off, as follows.

Suppose T_f were fixed. Nothing in (2) penalises the input directly, so the answer would be trivial: find whichever admissible u minimises β and stay there. Because T_f is itself controlled, the denominator of (24) rules that solution out. The operating point that aligns the indicator is in general not the operating point that prolongs life, so aligning now shortens the interval over which the alignment is averaged. The optimum must trade the two, and Section 6.5 measures how steep that trade is on a concrete plant. A fixed-horizon formulation represents none of it, which is why one was tried and discarded (Section 6.5).

5.1. A converse bound uniform over trajectories

Before computing anything we record a certificate that removes the conditioning. Theorem 4.1 bounds the loss along a trajectory the input has already chosen; the following proposition bounds it over all of them at once, and it is what would make the number computed below a limit rather than an observation, were a function satisfying it exhibited; none is, for reasons given below.

Proposition 5.1 (verification certificate). *Fix an indicator q and a constant $\lambda \geq 0$. Suppose there is a continuously differentiable V on a neighbourhood of the admissible state region such that*

- (i) $V(x) \leq 0$ for every x on the failure surface $\{\max_i x_i = 1\}$, and
- (ii) writing $D_i(x, u) = \varphi_i(u)h_i(x_i)$ for the drift of (2), for every admissible state x and every $u \in \mathcal{U}$,

$$\ell(\beta(x, u)) - \lambda + \nabla V(x)^\top D(x, u) \geq 0. \quad (25)$$

Then every admissible input satisfies $\bar{\ell} \geq \lambda + V(x_0)/T_f$. In particular, if $V(x_0) \geq 0$ then $\bar{\ell}^*(q) \geq \lambda$, uniformly over all admissible inputs and hence over all trajectories.

Proof. Let $u(\cdot)$ be any admissible input and x_t its trajectory. Since V is C^1 and $t \mapsto x_t$ is absolutely continuous, $t \mapsto V(x_t)$ is absolutely continuous and the chain rule gives $\frac{d}{dt}V(x_t) = \nabla V(x_t)^\top \dot{x}_t = \nabla V(x_t)^\top D(x_t, u_t)$ for almost

every t . Applying (25) at (x_t, u_t) gives $\ell(\beta(x_t, u_t)) - \lambda + \frac{d}{dt}V(x_t) \geq 0$ a.e.,
and integrating over $[0, T_f]$,

$$\int_0^{T_f} (\ell - \lambda) dt \geq V(x_0) - V(x_{T_f}) \geq V(x_0),$$

the last step by (i), since x_{T_f} lies on the failure surface. Dividing by $T_f > 0$
gives the claim. $\square$

Condition (25) constrains the plant data alone: it quantifies over every
state and every admissible input, and mentions no trajectory. Any V one
can exhibit therefore certifies a limit, and the search for a converse becomes
the search for a verification function.

The C^1 hypothesis is not a formality, and it is exactly what the nat-
ural candidates lack: the value function of the Dinkelbach problem be-
low is only Lipschitz and violates (25) pointwise, while affine functions are
smooth but certify only a vacuous pointwise bound. Supplementary Sec-
tion S3 records both attempts and what they imply, that a certificate must
be smooth and must have a gradient that varies, which points at a smooth
under-approximation of the value function by sum-of-squares programming
or interval arithmetic. We do not claim a certified converse here, and the
optimum computed below remains a computation; what Proposition 5.1 buys
is the replacement of an open problem without shape by a concrete object
with a checkable property.

Problems of the form (24) are handled by Dinkelbach’s transform [25]:
setting

$$F(\lambda) = \min_{u(\cdot) \in \mathcal{U}} \int_0^{T_f} (\ell - \lambda) dt, \quad (26)$$

the map F is decreasing in λ and vanishes exactly at the optimal ratio, so λ^* is
found by bisection on $F(\lambda) = 0$. The inner problem in (26) has free terminal
time and an absorbing set, hence a stationary Hamilton–Jacobi–Bellman
equation on the state alone, with no time argument. Because $\dot{x} > 0$ the
state advances monotonically, and a semi-Lagrangian scheme [26] converges
by value iteration. One implementation detail matters: the scheme must use
a fixed spatial step rather than a fixed time step, because degradation speeds
vary by orders of magnitude across $\mathcal{U}$ and a fixed time step is unusable at
one end of the envelope or the other.

707 6. Numerical study

708 Sections 3–5 are analytical, and three of their claims are stated sharply
 709 enough to be tested rather than illustrated: that the loss is invisible to a
 710 Fisher criterion, that the counting rule $m \geq d - 1$ binds in both directions on
 711 a generic plant, and that the best achievable loss is a finite number one can
 712 compute. This section instantiates the model on a synthetic three-mechanism
 713 plant on which every quantity is available in closed form, checks those three
 714 claims against it, measures what steering is worth, and asks which quantity,
 715 available in advance, says whether it will repay the effort. Section 7 then
 716 reports what measured fleets can and cannot establish about the premises.

717 6.1. The test plant

718 We take $d = 3$ mechanisms with Arrhenius factors $\varphi_i(u) = e^{\theta_i - E_i u}$, in
 719 which the input u is the reciprocal temperature $1/(kT)$, so that this is the
 720 Arrhenius form of Section 2 with e^{θ_i} the pre-exponential constant A_i . We
 721 set $\theta = (0, 0.3, -0.2)$ and $E = (0.6, 1.1, 0.8)$, and shapes $h_i(x) = 1 + c_i x$ with
 722 $c = (-0.5, 2.0, 0.8)$, giving one decelerating and two accelerating mechanisms.
 723 There is a single input $u \in [0, 3]$, so $m = 1 < d - 1 = 2$, and Theorem 4.1 puts
 724 zero loss out of reach unless this plant is one of the exceptions of Remark 4.5;
 725 Section 6.3 computes the floor and finds that it is not. Noise is unit and the
 726 prior variance is $\text{Var}[R] = 10^3$, so the weight $w = \gamma/(1 + \gamma)$ of Lemma 2.2
 727 is evaluated at the plant’s own $\gamma = \|\mathbf{d}\|^2 \text{Var}[R]$ at each state rather than
 728 held fixed; γ varies along a trajectory because $\|\mathbf{d}\|$ does. Every trajectory
 729 starts from $x_i(0) = 5 \times 10^{-3}$ and runs to the condemnation limit $x_f = 0.9$ of
 730 (3). Raising x_f to unity leaves every qualitative conclusion below intact and
 731 moves the constants: the peak floor of Section 6.3 rises from 4.52° to 5.41° ,
 732 and the gain of Section 6.5 falls from 68 to 58. No constant reported below
 733 should be read across thresholds.

734 *Verification of the identities.* The analytical identities were checked numeri-
 735 cally before use, and the residuals are reported in Supplementary Section S4.

736 6.2. A Fisher level set of varying loss

737 Proposition 4.6 is easy to demonstrate on this plant. For a constant input,
 738 a common gain on $\mathbf{d}$ moves $\|\mathbf{d}\|$ while leaving $\hat{\mathbf{d}}(\tau)$, and hence β , untouched:
 739 the Fisher score and the loss can be varied independently. Write $\delta(u)$ for the
 740 gain that equalises the score across inputs, fixed by $\delta(u)^2 \overline{\|\mathbf{d}\|^2}(u) = F$ with

Table 2: Loss of the fixed indicator at a Fisher score held equal. The exposure factor δ , defined in the text, holds $\delta^2 \|\mathbf{d}\|^2$ fixed and leaves $\hat{\mathbf{d}}(\tau)$ unaffected. Losses use $\gamma = \|\mathbf{d}\|^2 \text{Var}[R]$ with $\text{Var}[R] = 10^3$.

u	δ	$\delta^2 \ \mathbf{d}\ ^2$	rms β (deg)	$\bar{\ell}$ (nat)
0.0	0.235	0.4307565	6.23	0.0215
1.0	0.628	0.4307565	4.53	0.0123
1.5	1.000	0.4307565	6.22	0.0208
2.0	1.564	0.4307565	7.53	0.0379
3.0	3.769	0.4307565	6.57	0.1161

741 F its value at $u = 1.5$. It is realised by rescaling how long the component
742 runs under load, which is why it does not rotate $\hat{\mathbf{d}}$. By construction $\delta = 1$
743 at $u = 1.5$, and $\delta > 1$ wherever the raw score is smaller, so it is a multiplier
744 on exposure rather than a fraction of it. Table 2 does exactly that, holding
745 the Fisher score equal to seven digits by tuning the exposure factor δ . The
746 loss still spans a factor 9.4.

747 The two objectives also disagree on which input to prefer: the raw score,
748 before δ equalises it, is largest at $u = 0$, which is why δ is smallest there, and
749 the loss is 1.7 times its value at $u = 1$. How far they disagree is a property
750 of the instance; what is structural is only the blindness.

751 6.3. The reachability floor with a single input

752 With $m = 1 < d - 1$ no direction lies on every reachable set at once. Along
753 the closed loop at q^* the floor is largest at the start of life, where $\beta_{\min} =$
754 4.52° ; it falls through mid-life to below the 0.01° resolution of our sampling,
755 where the indicator is momentarily reachable, and rises again towards failure
756 (Fig. 5(a)). Zero average loss would need the indicator reachable at every
757 instant and not at one, so it is unattainable however the input is steered, in
758 agreement with Theorem 4.1. Fig. 4(a) shows the geometry: the reachable
759 set sweeps across q^* as the plant ages, so the mid-life arc passes close to the
760 indicator while the arcs at $\tau = 0.05$ and $\tau = 0.9$ stand off it.

761 6.4. The counting rule, tested in both directions

762 Theorem 4.1 is an equivalence, and the paragraph above tests only half
763 of it: with too few inputs the floor stays positive. The other half, that
764 supplying the inputs the counting rule asks for makes the floor vanish, is the

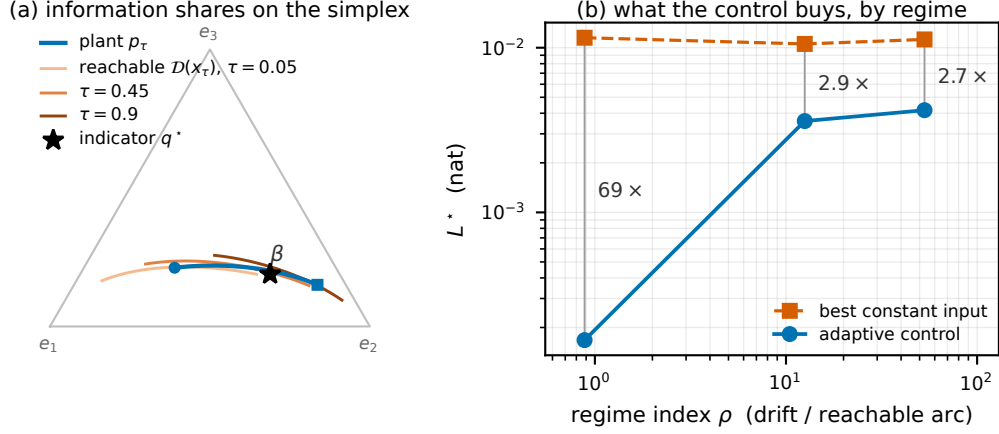


Figure 4: (a) The information-share curve p_τ on the 2-simplex (circle: start, square: failure), the reachable sets $\mathcal{D}(x_\tau)$ at $\tau = 0.05, 0.45$ and 0.9 , and the optimal indicator q^* . No point lies on all three, so β cannot be held at zero (Theorem 4.1). (b) What steering buys, against ρ , the drift of the reachable arc over its length.

more useful claim for a practitioner deciding how many operating variables to instrument, so we test it directly.

Give the same three-mechanism plant a second independent input by writing $\varphi_i(u) = \exp(\theta_i - E_i u_1 - G_i u_2)$ with $G = (0.9, 0.3, 1.2)$, so that $m = 2 = d - 1$ and each input acts on the mechanisms in a different pattern. Then $\log \mathbf{d}_i$ is affine in (u_1, u_2) , and matching a target direction fixes the $d - 1 = 2$ differences of logarithms: two linear equations in two unknowns, with a coefficient matrix of determinant 0.27 here, hence a unique solution at every state.

Taking as target the direction the plant occupies at mid-life under a central operating point, that solution lies inside the admissible box $[0, 3]^2$ at every one of 600 sampled states covering the whole life, and the residual misalignment there is 8.5×10^{-7} degrees, that is, zero to numerical precision; the schedule it requires is drawn in Fig. 5(b). An independent 90×90 grid search over the box returns a floor of at most 0.156° , which is the resolution of the grid rather than a property of the plant.

The contrast with the single-input case is the content of the theorem: the same mechanisms, the same shapes, the same envelope on each input, and the peak floor moves from 4.52° to zero at every sampled state on the addition of one operating variable. On this plant the counting rule is therefore not

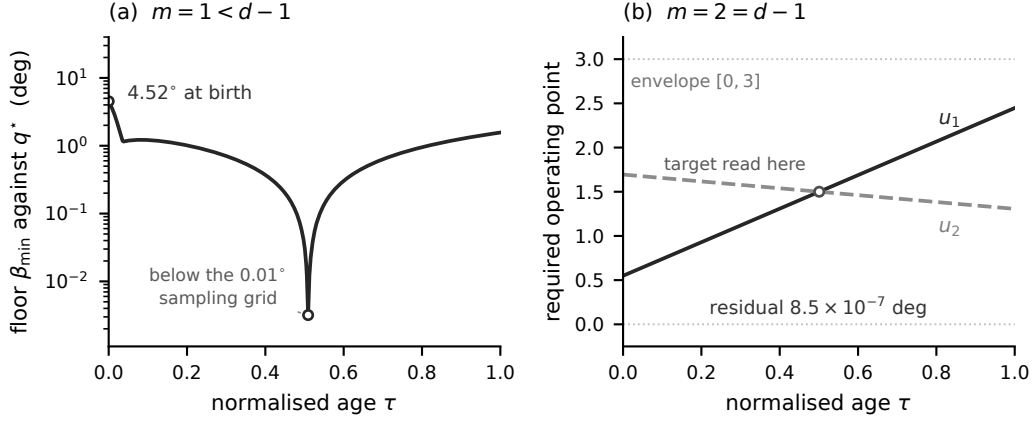


Figure 5: (a) With $m = 1$ the floor against q^* is largest at birth and dips below the sampling grid near mid-life. (b) With $m = 2$ the mid-life direction is held exactly; plotted is the operating point this requires, which never leaves $\mathcal{U}$.

785 a conservative sufficient condition but a sharp one, and within the adopted
 786 model it fixes how many independent control degrees of freedom the plant
 787 must offer. How those degrees of freedom correspond to physical actuators,
 788 and how the mechanisms are individuated in the first place, are modelling
 789 questions this paper does not settle. Proposition 4.4 says which plants escape
 790 it, and Remark 4.5 says why a designer should not count on being one of
 791 them.

792 6.5. The value of steering

793 Minimising over q gives $q^* = (0.466, 0.771, 0.435)$, at which the adaptive
 794 policy attains 1.68×10^{-4} against 1.15×10^{-2} under the best constant input,
 795 a factor of 68. The ratio depends on how finely $\mathcal{U}$ is discretised for the policy,
 796 rising to 72 at 161 points; we report it on the 41-point grid used throughout.

797 The factor is a ratio of information losses, not of information, and it is
 798 not free in reliability terms: the policy that attains it runs the component
 799 differently, and Section 6.6 finds the two closed loops separated by a fac-
 800 tor of 3.4 in life. Aligning the indicator and prolonging the component are
 801 competing objectives, which is what makes the horizon in (24) a controlled
 802 quantity rather than a constant. That factor is the value of steering at all,
 803 not the value of steering optimally: a myopic policy that attends only to the
 804 numerator of (24) sits 2.8% above the optimum at q^* .

805 The fraction in (24) is doing visible work at this operating point. The
806 closed loop at q^* and the closed loop at a balanced indicator, which calls for
807 far less steering, differ by a factor 3.4 in lifetime: what alignment buys in the
808 numerator is paid for in the denominator. This is the tension anticipated in
809 Section 5, and it is why the horizon cannot be fixed in advance.

810 6.6. *The computed fractional optimum*

811 Dinkelbach bisection with the control grid matched to the policy gives,
812 over $N \in \{25, 29, 35\}$, extrapolants agreeing to 1.2% and $\lambda_\infty^* = 1.63 \times 10^{-4}$,
813 below the realisable 1.68×10^{-4} as it must be.

814 We quote this to two digits only, for the following reason: the raw values
815 are twice the extrapolant, so the first-order correction carries half the answer,
816 and both quantities still move by 4% between control grids of 41 and 81
817 points. Three checks guard the number: F must decrease and cross zero, λ^*
818 must not exceed a realisable policy, and the grid sequence must extrapolate
819 consistently. A fixed-horizon formulation failed the second and third and was
820 discarded, as did a first attempt with mismatched control grids. A converse
821 bound uniform over trajectories, which would replace this computation by a
822 theorem, remains open.

823 6.7. *Authority, drift, and how far they screen*

824 For a practitioner the first question is not how to steer but whether to
825 bother. The reachable set $\mathcal{D}(x)$ is an arc, and it offers two natural answers:
826 its mean length, which is the actuation authority available to chase the in-
827 formative direction, and the drift of its midpoint over the life, which is how
828 far that direction wanders. Write ρ for the drift divided by the length, which
829 weighs the two against each other.

830 Table 3 varies the spread of the activation energies, which by (20) is what
831 sets that authority. As the arc collapses, the attainable loss L^* rises by a
832 factor 25 and the value of control falls from 68 to 2.7: with weak authority
833 there is little left to steer. Fig. 4(b) plots the two against ρ . In this family the
834 drift barely moves, staying between 24° and 30° while the arc spans nearly
835 two decades, so ρ is the reciprocal of the arc in all but name and the table
836 cannot say which of the two matters. Both can be computed from a nominal
837 degradation model before any indicator is designed.

838 Three plants differing only in their activation energies are not a screening
839 law, so the index was tested on 299 further plants drawn at random: two to

840 five mechanisms, one or two independently adjustable inputs, and the pre-
 841 exponentials, activation energies, shape coefficients, input envelope, initial
 842 state and signal-to-noise ratio varied together. The rank correlation between
 843 ρ and the gain is -0.36 , with a bootstrap 95% interval from -0.46 to -0.24 ,
 844 and it keeps its sign inside every stratum of the mechanism count and of the
 845 input count, so it is not an artefact of either.

846 Read as a classifier of whether steering repays a factor of five, which 32%
 847 of the plants do, ρ has an area under the receiver-operating curve of 0.73,
 848 with interval 0.67 to 0.79. The reachable arc alone scores 0.73 as well, and
 849 the drift alone 0.53, with interval 0.46 to 0.60, which includes the 0.5 of a
 850 coin toss.

851 Equal areas do not make the two equally good, and the question is settled
 852 by a model that contains both as special cases: a logistic regression on the
 853 logarithms of arc and drift as separate terms, of which the arc alone and the
 854 ratio ρ are each a constrained case. The drift adds nothing to the arc as
 855 a main effect ($p = 0.47$), and constraining the two to enter as a ratio fits
 856 significantly worse than the arc alone ($p = 7.7 \times 10^{-5}$; Akaike information
 857 351 against 336). Both results hold in each input count separately. The
 858 drift does carry information, but through the arc ($p = 3.0 \times 10^{-4}$ for their
 859 interaction), and in opposite directions at the two ends. Among plants of
 860 low authority the share repaying fivefold falls from 0.18 to 0.08 as the drift
 861 rises from its lowest to its highest third; among plants of high authority it
 862 rises from 0.48 to 0.61. Drift that outruns the authority defeats steering, and
 863 drift that the authority can follow is what gives steering something to do.
 864 The ratio encodes only the first of these. It is also why the lowest tenth of
 865 ρ repays fivefold only 43% of the time: there the drift is small, a median of
 866 2.7° against 14° over the sample.

867 Out of sample the ranking agrees. Over repeated ten-fold cross-validation
 868 the arc and ρ each rank unseen plants with an area of 0.72, and a model
 869 adding the drift and its interaction with the arc reaches 0.74, higher in every
 870 repeat. The quantity to screen with is therefore the arc, the authority itself,
 871 refined by the drift only through that interaction; ρ remains a convenient
 872 description of the family of Table 3, where the drift is nearly fixed. The first
 873 row of that table is not typical of its own regime: among sampled plants
 874 within a factor 1.4 of its ρ , the median gain is 5.2 rather than 68.

Table 3: Authority and drift in a one-parameter family. The activation energies E set the actuation authority; c , the noise and $\mathcal{U}$ are held fixed. L^* is the loss under adaptive control, and the gain its ratio to the best constant input.

E	arc	ρ	L^*	gain
(0.60, 1.10, 0.80)	33.9°	0.88	1.7×10^{-4}	68×
(0.80, 0.82, 0.79)	1.93°	12.5	3.6×10^{-3}	2.9×
(0.80, 0.805, 0.798)	0.45°	53.1	4.2×10^{-3}	2.7×

875 6.8. What the loss buys a downstream estimator

876 Corollary 2.4 prices the loss in the error of the estimate the model itself
877 prescribes. A practitioner does not use that estimate, so the ordering is worth
878 checking against a procedure that uses none of the construction: sample the
879 indicator over the last 40% of life, match the observed history against the
880 indicator’s own reference curve, and read remaining life off the best-fitting
881 time offset. That estimator sees a scalar signal and nothing else — not $\hat{\mathbf{d}}$,
882 not γ , not the loss.

883 Over twelve indicators spanning $\bar{\ell}$ from 0.012 to 0.37, its median absolute
884 error — one of the standard prognostic accuracy measures [27] — rises with
885 the loss, at Spearman correlation 0.89, and the best-aligned indicator beats
886 the worst by a factor 1.5. Twelve indicators is a small sample, so both
887 are given with intervals obtained by resampling the 160 units behind each
888 median: 0.34 to 0.90 for the correlation, and 1.40 to 2.03 for the factor. The
889 ordering is not in doubt — a one-sided permutation test gives $p = 1.5 \times 10^{-4}$
890 — but its strength is, and the point estimate sits near the top of its own
891 interval. The factor is far smaller than the 31-fold spread in $\bar{\ell}$, and it should
892 be: the loss is a logarithmic information quantity, the error is a time, and
893 an estimator with the whole history recovers much of what any one instant
894 discards.

895 The same protocol on 120 further plants, drawn at random as in Sec-
896 tion 6.7, gives correlations between 0.87 and 0.99, with the plant above
897 among the weaker. That design tilts every indicator toward the same di-
898 rection, however, so loss and error both grow with one parameter and their
899 agreement is close to structural. Tilting instead by a random amount of
900 up to 60° in a random direction weakens the median correlation to 0.74,
901 with the interval excluding zero on 84% of plants. It also exposes what the
902 experiment cannot show. On these plants the angle between an indicator and
903 the mid-life informative direction ranks the indicators identically to the loss

on 82% of plants, with neither consistently better on the rest, because the direction rotates little beside how far the indicators differ. The experiment is therefore evidence that the ordering of the loss carries over to an estimator built from none of this construction; it is not evidence that the time-averaged loss improves on a mid-life snapshot of the misalignment, which would need a plant whose direction rotates by more than its candidate indicators differ.

7. What public data can say about the premises

The construction rests on three modelling choices, and each is empirically testable. They do not carry equal weight, and Fig. 6 separates what depends on which. Two are premises about the plant: that the informative direction rotates along the trajectory, and that the input moves it. The third is the separability of Lemma 4.3, which is what makes the operator’s authority computable offline. Table 4 summarises what six public run-to-failure fleets currently settle. Supplementary Section S1 gives the protocol behind every row, and its Table S1 lists the parameters of the code that produced it.

Three readings qualify the table. The Severson channels are capacity fade, resistance growth and mean temperature, and mean temperature is a monitoring channel rather than a damage state — it is the operating variable itself in the Arrhenius form of Section 2. That row therefore reads the rotation of a direction formed from three measured series, which is what the premise asserts, and does not need (5)’s pairing of channels with mechanisms. On the four measured fleets the obstacle to testing actuation is the estimator and not the physics, so the premise is supported in simulation and untested on hardware. Both the Severson and the C-MAPSS rows estimate the direction by differentiating each channel along normalised life, which reads a time derivative as the sensitivity of Eq. (6) and so presumes the time-shift population of Section 2. And the C-MAPSS contrast is reported against a negative control — fake regimes cut from a single real one, where no contrast exists by construction — because a procedure that returned a positive contrast regardless would produce the same table entry.

The consequence is the scope of the results above. The geometry of Sections 3 and 4 needs only $\varphi_i > 0$ and $h_i > 0$, and is untouched. What separability buys, and what the rejection withdraws, is Proposition 4.4 and the offline route to $\mathcal{B}$: those describe a model class, and a plant must be shown to belong to it before they are used. The two-way layout above is the test, and it costs one pass over data a reliability programme already holds.

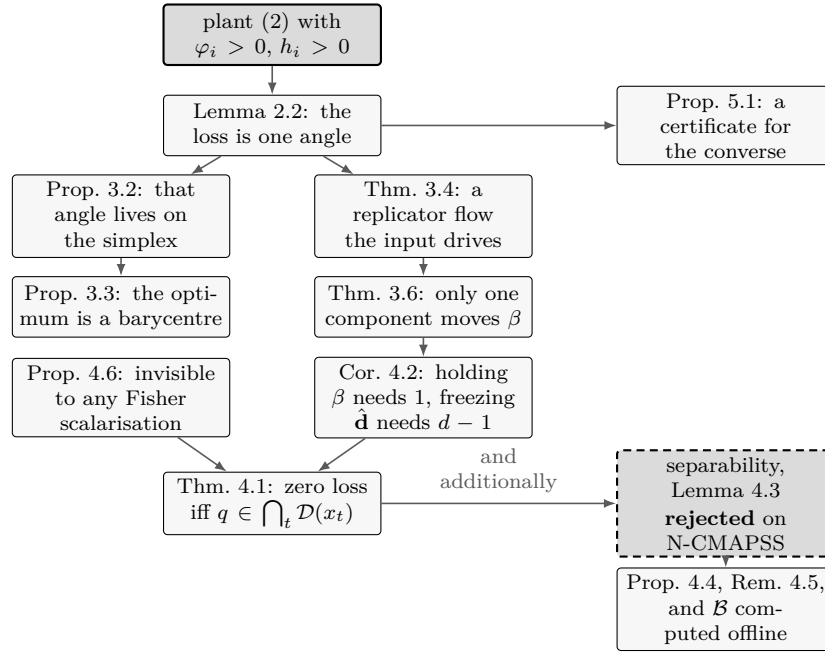


Figure 6: What rests on what. The spine needs only the positivity of the degradation model. Only the dashed branch adds separability, which Section 7 reports as rejected on the best public turbofan data.

940 8. Discussion

941 Four practical readings follow from the results above. The first two hold
942 whatever the plant is, because one is a test and the other an invariance. The
943 last two assume the separable form of (2), which Section 7 reports as rejected
944 on the only data able to test it, so they apply where that form does and not
945 otherwise.

946 Test the separability assumption before relying on it. The two-way layout
947 of Section 7 needs only a fleet whose operating point varies within a unit,
948 and costs one pass over data a reliability programme already holds. If the
949 interaction clears the measured noise floor, the offline route to the operator's
950 authority is unavailable and the two model-based readings below do not
951 apply.

952 No Fisher criterion can rank fixed indicators by the loss they incur. This
953 is the second of the two unconditional readings: Proposition 4.6 is an in-
954 variance of the loss under a common gain and holds whatever the plant, so
955 rules that maximise a scalarisation of the Fisher information cannot separate

Table 4: What the six public fleets settle, and what they cannot. Only the two turbofan simulations vary the operating point within a unit, which is why the actuation premise is supported in simulation and untested on measured hardware.

fleet	premise	what is measured	reading
Severson lithium-ion [28]	rotation	arc of an empirically constructed informative direction, the fleet mean, over $\tau \in [0.15, 0.9]$, on 129 Severson lithium-ion cells	travels 30.7° of arc and ends 21° from its start (Fig. 2(c)); stays in the open positive orthant once the window reaches 301 cycles
XJTU-SY [29], PRONOS-TIA [30]	actuation	—	the operating condition is fixed per unit, so condition and group of units cannot be separated
NASA batteries [31]	actuation	—	four cells
C-MAPSS [32]	actuation	between-regime minus within-regime angle, each engine serving as its own control	by 7.9° (FD004) and 10.8° (FD002), positive for 436 of 439 engines; fake regimes give $ B - W \leq 0.5^\circ$
N-CMAPSS DS02, DS01 [33]	separability	interaction against both main effects, in a two-way layout inside engines	larger by a factor 1.65 over the operating-point effect on DS02 and 1.84 on DS01, under every clustering and staging tried: rejected

956 configurations whose losses differ by an order of magnitude. Where the two
957 objectives do coincide, that is a property of the application, to be verified
958 rather than assumed.

959 The binding constraint is actuation authority, not indicator sophistica-
960 tion. Theorem 4.1 prices the operating variables a lossless fixed indicator
961 would need, and Proposition 4.4 sharpens the count to sufficient but not
962 necessary, with exceptions that Remark 4.5 shows to be a knife edge. Care
963 in choosing the weights substitutes for actuation in neither case. What one
964 controllable variable does buy is the weaker objective of Corollary 4.2: an
965 installation that could never freeze the informative direction can still hold
966 the misalignment constant, though not at zero. Authority bought this way
967 is not free, and Section 6.6 prices it in life.

968 The decision to steer precedes the design of the indicator. The width of
969 the reachable arc requires only a nominal degradation model and the operat-
970 ing envelope, and Table 3 shows the value of steering collapsing from $68\times$ to
971 under $3\times$ as it shortens. It screens rather than decides, and across random

972 plants the drift of the informative direction refines it only in combination:
 973 drift the authority can follow gives steering something to do, and drift that
 974 outruns it defeats steering (Section 6.7).

975 *Limitations.* Four limitations qualify the readings above. The separable form
 976 of (2) is not only an idealisation but one the data reject: Section 7 finds
 977 the interaction between operating point and state larger than either main
 978 effect on two N-CMAPSS datasets, so computing the operator’s authority
 979 offline is available only where that form holds. The optimum of Section 5
 980 is computed rather than certified, to two digits. The actuation premise is
 981 supported by simulated rather than measured data, for the reasons set out in
 982 Section 7. And the population is taken to slide along one degradation path,
 983 which is what makes remaining life a coordinate on it; under dispersion in
 984 the coefficients of (2) the sensitivity that Remark 2.1 measures stands some
 985 45° from (6), so the construction describes fleets that vary in when they fail
 986 rather than in how they degrade.

987 9. Conclusion

988 A fixed health indicator discards information, and this paper has treated
 989 what it discards as a quantity the operating point controls. Indicator design
 990 then becomes a reachability question. The family of reachable direction sets
 991 bounds what the operating point can achieve, and their width, available
 992 before any indicator is fixed, indicates whether substantial benefit is plau-
 993 sible, and across random plants of this model the drift of the informative
 994 direction refines that indication only in combination with it. Under the pro-
 995 posed model, actuation authority, rather than sophistication of the indicator,
 996 governs whether a fixed compression can come close to lossless. Whether
 997 that authority is present is an empirical question rather than a modelling
 998 assumption, and on 439 simulated turbofan records the evidence supports
 999 it. Establishing the same on measured fleets awaits records in which the
 1000 operating point varies enough to show the effect.

1001 Section 7 rejects that separable form on the highest-fidelity public tur-
 1002 bofan data, so the geometry of Section 4.1, and the offline route it opens,
 1003 describe a model class rather than measured machinery. What replaces that
 1004 structure when the stress model is not separable, and whether a weaker one
 1005 still yields a computable reachability criterion, remains open.

Two further questions remain open, the first of them now in a more specific form. Proposition 5.1 reduces the search for a converse bound uniform over trajectories to the exhibition of a single continuously differentiable verification function; the value function is the natural candidate and fails the differential test for a structural reason, so what is required is a smooth under-approximation of it, by sum-of-squares programming or by interval arithmetic with an explicit Lipschitz constant. The other is unchanged: the computed optimum to better than two digits, which requires a scheme of higher order than the semi-Lagrangian one used here.

Data availability and reproducibility

The six fleets named in Section 7 are publicly available. One script recomputes every reported number from a single configuration and compares it with the value written in this manuscript and in its supplementary material, covering the tables and the quantities reported alongside them; all of its checks pass, and the output of the run that verified this version is included. The random-plant sweep of Section 6.7 takes hours, so it and the downstream repeat of Section 6.8 are recorded by two further scripts, and the checks read the records. Three groups of numbers lie outside the checks: the residuals of the identity checks in Supplementary Section S4, the barycentre comparison in Section 3.3, and the diagnostics of the two candidate certificates in Supplementary Section S3. The plots are produced by three further scripts. All of these are available at <https://github.com/HuyHoangLe0201/health-indicator-information-loss>.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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